Vaccination Cohort & Dose Scheduling System

Requirements Engineering & UML Use-Case Modeling

Problem Statement: Healthcare & Telemedicine

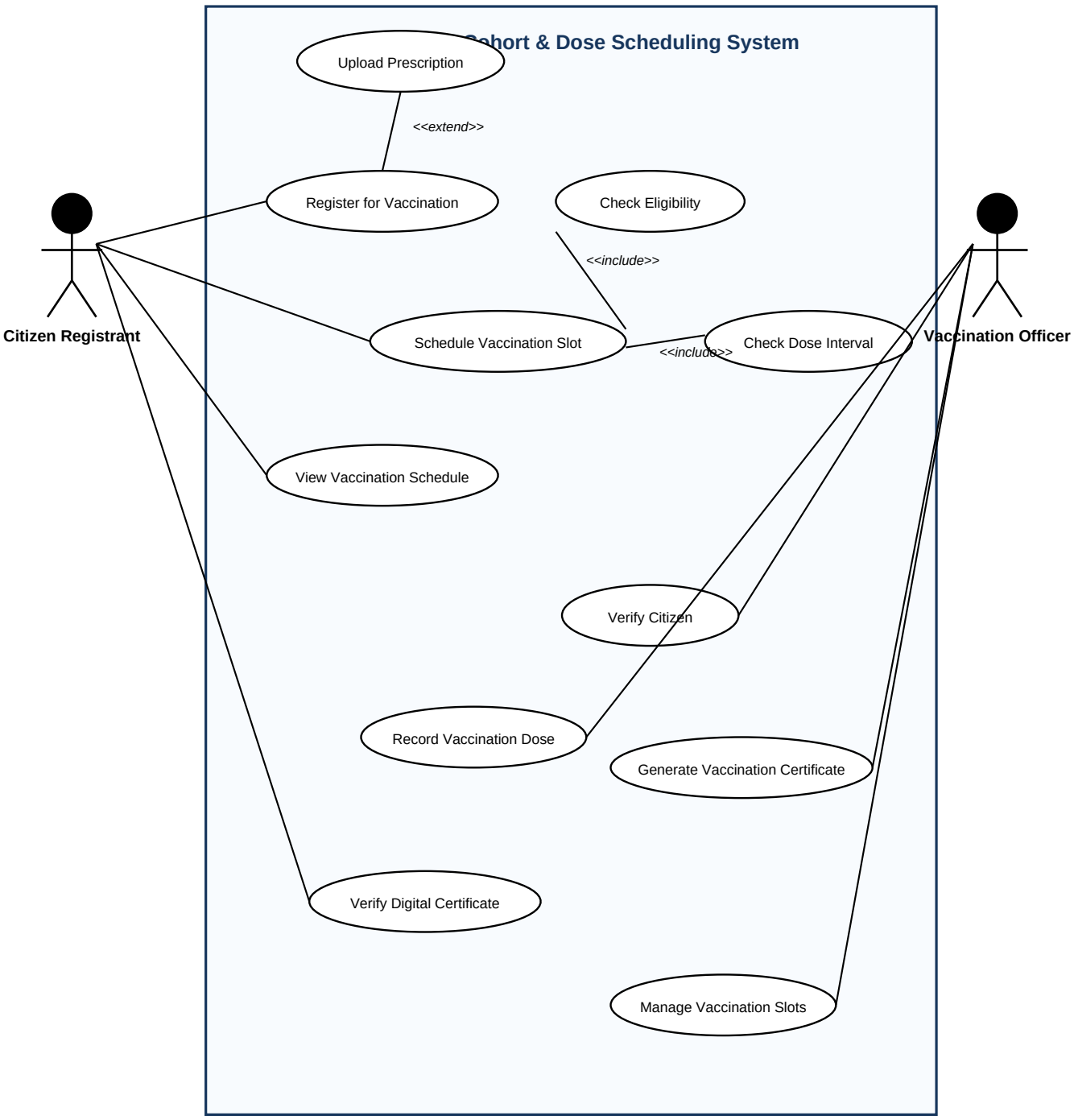
A public-health administration platform for organizing demographic vaccination rollouts, enforcing minimum interval days between doses, scheduling center slots, and generating verifiable QR digital vaccination certificates.

1. Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enforce the minimum required interval between vaccine doses before allowing a subsequent dose to be scheduled.	High	If the required interval is 28 days and fewer than 28 days have passed since Dose 1, the system shall block Dose 2 booking. If 28 or more days have passed, the system shall allow booking.	Prevents citizens from receiving subsequent doses earlier than the required medical interval.
FR-002	Functional	The system shall allow a Citizen Registrant to register for vaccination by providing the required demographic and vaccination information.	High	A citizen providing all mandatory information shall be successfully registered. Registration shall be rejected if mandatory information is missing or invalid.	Provides the citizen information required for eligibility checking and vaccination scheduling.
FR-003	Functional	The system shall allow an eligible Citizen Registrant to view available vaccination-center slots and book a suitable slot.	High	The system shall display available slots to an eligible citizen. Once a slot is successfully booked, it shall no longer be available for another booking.	Enables organized vaccination scheduling and prevents double booking of available capacity.
FR-004	Functional	The system shall allow a Vaccination Officer to record the administration of a vaccine dose for a registered citizen.	High	After verifying the citizen, the officer shall be able to record the vaccine, dose number, date, and vaccination center. The system shall save the vaccination record successfully.	Maintains an accurate vaccination history and supports subsequent dose scheduling.
FR-005	Functional	The system shall generate a digitally verifiable vaccination certificate after a vaccine dose has been successfully recorded.	High	After successful dose recording, the system shall generate a certificate containing required vaccination details and a digitally verifiable QR code.	Provides reliable proof of vaccination and enables certificate verification.
FR-001	Non-Functional — Performance & Security	The vaccination certificate verification endpoint shall authenticate digitally signed QR codes and respond within 150 ms under the specified simulated peak load.	High	Under the specified simulated peak load, valid digitally signed QR codes shall be authenticated and the verification response shall be returned within 150 ms. Invalid or tampered QR codes shall fail verification.	Ensures certificate verification is secure and fast enough for practica use.
FR-002	Non-Functional — Availability & Reliability	The system shall reliably maintain vaccination and appointment information and remain available during vaccination campaigns.	High	Authorized users shall be able to access the system during normal operating periods, and confirmed vaccination and appointment records shall remain intact after normal system failures.	Vaccination scheduling and record are time-sensitive, so the system must be dependable and prevent loss of critical information.

2. UML Use-Case Diagram

The model contains the two actors specified in the problem statement and includes both required <<include>> and <<extend>> relationships.



Relationship note: Schedule Vaccination Slot includes Check Eligibility and Check Dose Interval because both are mandatory before booking. Upload Prescription extends Register for Vaccination because it is conditional.

3. Main Success Scenario & Alternate Flow

Core Use Case: Schedule Vaccination Slot

Primary Actor: Citizen Registrant

Goal: Book an appropriate vaccination-center slot while enforcing eligibility and the minimum interval between doses.

3.1 Preconditions

1. The Citizen Registrant is registered in the system.
2. Required demographic and vaccination information is available.
3. The citizen is eligible for the vaccination being scheduled.
4. For a subsequent dose, the previous dose information is available.
5. At least one applicable vaccination-center slot is available.

3.2 Main Success Scenario

1. The citizen selects **Schedule Vaccination Slot**.
2. The system retrieves the citizen's registration and vaccination history.
3. The system checks the citizen's vaccination eligibility.
4. The system checks the minimum required interval since the previous dose, if applicable.
5. The system displays available vaccination-center slots.
6. The citizen selects a preferred available slot.
7. The system validates that the selected slot is still available.
8. The system confirms and records the appointment.
9. The system updates the citizen's vaccination schedule and displays the appointment details.

3.3 Postconditions

1. A vaccination-center slot is reserved for the citizen.
2. The appointment is recorded in the system.
3. The selected slot is no longer available for another booking.
4. The citizen's vaccination schedule reflects the new appointment.

3.4 Alternate Flow — Minimum Dose Interval Not Satisfied

A1.1 The system retrieves the date of the citizen's previous dose.

A1.2 The system determines that the minimum required interval has not elapsed.

A1.3 The system prevents the subsequent-dose booking.

A1.4 The system informs the citizen that the dose is not yet eligible for scheduling.

A1.5 The system displays the earliest eligible date for the next dose.

Traceability: This use case directly realizes FR-001 (dose interval enforcement) and FR-003 (vaccination slot scheduling).